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**Batch – 15**

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## 1. Algorithm

1. Start.
2. Declare integer variables `m`, `n`, `choice`, `i`, and `j`.
3. Accept values of `m` and `n`.
4. Check whether:
  - `m > 0`
  - `n > 0`
  - `m <= n`
5. If the input is invalid, display an error message and ask for `m` and `n` again.
6. Display the menu:
  - Multiplication tables of `m` and `n`
  - Multiplication grid
  - Consecutive integers
  - Exit
7. Accept `choice`.
8. Use a `switch` statement:
  - **Case 1:** Use a loop from 1 to 10 and display the multiplication tables of `m` and `n` side by side.
  - **Case 2:** Use nested loops. The outer loop runs from 1 to `m`, and the inner loop runs from 1 to `n`. Display `i * j` to form the multiplication grid.
  - **Case 3:** For every integer `i` from `m` to `n`, use an inner loop from 0 to 5 and display `i + j`. This prints the integer itself followed by the next five consecutive integers.
  - **Case 4:** Display an exit message and terminate the menu loop.
  - **Default:** Display an error message for an invalid menu choice.
9. Repeat steps 6–8 until the user selects option 4.
10. Stop.

## 2. Test-Case Table

| Test Case | Input(s)           | Expected Output(s)                                      | Actual Output(s)           | Result |
|-----------|--------------------|---------------------------------------------------------|----------------------------|--------|
| 1 – Valid | m=3, n=5, Choice 1 | Tables of 3 and 5 from ×1 to ×10 displayed side by side | Tables displayed correctly | Pass   |
| 2 – Valid | m=3, n=5, Choice 2 | 3 × 5 multiplication                                    | Correct 3 × 5 grid         | Pass   |

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|              |                    |                                                    |                                        |      |
|--------------|--------------------|----------------------------------------------------|----------------------------------------|------|
| 3 – Valid    | m=3, n=5, Choice 3 | 3: 3 4 5 6 7 8, 4: 4 5 6 7 8<br>9, 5: 5 6 7 8 9 10 | Correct consecutive integers displayed | Pass |
| 4 – Boundary | m=1, n=1, Choice 1 | Table of 1 displayed in both columns               | Correct output                         | Pass |
| 5 – Boundary | m=1, n=1, Choice 2 | Grid containing only 1                             | Correct output                         | Pass |
| 6 – Invalid  | m=0, n=5           | Error because m is not positive                    | Error message displayed                | Pass |
| 7 – Invalid  | m=5, n=3           | Error because m > n                                | Error message displayed                | Pass |
| 8 – Invalid  | m=3, n=5, Choice 9 | Invalid menu choice message                        | Error message displayed                | Pass |
| 9 – Exit     | m=3, n=5, Choice 4 | Program exits                                      | Exiting program.                       | Pass |

## 3. C PROGRAM

```
#include <stdio.h>
int main() {
    int m, n, choice, i, j;
    /* Input and validation of m and n */ do {
        printf("Enter two positive integers m and n (m <= n): ");
        scanf("%d %d", &m, &n); if (m <= 0 || n <= 0 || m > n) {
            printf("Invalid input. Enter positive integers with m <= n.\n");
        } while (m <= 0 || n <= 0 || m > n);
    } /* Menu */
    do {
        printf("\n1. Multiplication tables of m and n\n");
        printf("2. Multiplication grid\n");
        printf("3. Consecutive integers\n");
        printf("4. Consecutive Integers from %d to %d\n", m, n);
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                for (i = 1; i <= n; i++) {
                    for (j = 1; j <= m; j++) {
                        printf("%d\t", i * j);
                    }
                    printf("\n");
                }
                break;
            case 2:
                for (i = 1; i <= n; i++) {
                    for (j = 1; j <= m; j++) {
                        printf("%d\t", i * j);
                    }
                    printf("\n");
                }
                break;
            case 3:
                for (i = m; i <= n; i++) {
                    printf("%d\t", i);
                }
                printf("\n");
                break;
            case 4:
                return 0;
            case 5:
                return 0;
            default:
                printf("Invalid choice\n");
                break;
        }
    } while (choice != 4 & choice != 5);
}
```

```
"(m=%d, n=%d)\n\n",
m, n, m, n); for (i =
m; i <= n; i++)
{ printf("%d:", i); for
(j = 0; j <= 5; j++)
{
printf(" %d", i + j);
}
}
break;
printf("\n"); case 4:
break;
printf("\nExitdefault:
printf("\nInvalid menu choice. "
"Please enter a number from 1 to
4.\n");
}
} while (choice != 4);
return 0; }
```

## 4. Execution

### Choice 1

Enter two positive integers m and n ( $m \leq n$ ): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 1

Multiplication Tables of 3 and 5 ( $m=3$ ,  $n=5$ )

|             |             |
|-------------|-------------|
| 3 x 1 = 3   | 5 x 1 = 5   |
| 3 x 2 = 6   | 5 x 2 = 10  |
| 3 x 3 = 9   | 5 x 3 = 15  |
| 3 x 4 = 12  | 5 x 4 = 20  |
| 3 x 5 = 15  | 5 x 5 = 25  |
| 3 x 6 = 18  | 5 x 6 = 30  |
| 3 x 7 = 21  | 5 x 7 = 35  |
| 3 x 8 = 24  | 5 x 8 = 40  |
| 3 x 9 = 27  | 5 x 9 = 45  |
| 3 x 10 = 30 | 5 x 10 = 50 |

## Choice 2

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 2

Multiplication Grid (m=3, n=5)

|   |  | 1     | 2 | 3 | 4  | 5  |
|---|--|-------|---|---|----|----|
|   |  | ----- |   |   |    |    |
| 1 |  | 1     | 2 | 3 | 4  | 5  |
| 2 |  | 2     | 4 | 6 | 8  | 10 |
| 3 |  | 3     | 6 | 9 | 12 | 15 |

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice:

## Choice 3

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 3

Consecutive Integers from 3 to 5 (m=3, n=5)

3: 3 4 5 6 7 8

4: 4 5 6 7 8 9

5: 5 6 7 8 9 10

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: █

**Invalid Choice**

Enter two positive integers m and n ( $m \leq n$ ): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 9

Invalid menu choice. Please enter a number from 1 to 4.

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: █